WAREHOUSE & CLIENT INVENTORY MANAGEMENT SYSTEM

System Design Specification

Status	Design only — no implementation started
Version	1.0 — consolidated design baseline
Prepared	13 October 2026
Scope	Company warehouses, client inventory, manufacturing, department issues and returns

Hosting selection is intentionally deferred until operational requirements are reviewed.

1. Purpose, status, and scope

This is the consolidated design baseline for a warehouse, distribution, client inventory, manufacturing, and department issue/return system. STATUS: DESIGN ONLY — no implementation or code has started. Hosting/provider selection remains deferred.

Scope includes multiple company warehouses and clients; warehouse–client assignments; company and client catalogs; stock-in/out; transfers; shipments and receipts; opening balances; corrections; returns and write-offs; manufacturing; department issues; approvals; reservations; batch/expiry tracking; bulk imports; audit; notifications; and reporting.

Online-only operation. Inventory valuation, purchase cost, selling price, and currency fields are out of scope. Normative terms: MUST = required; SHOULD = recommended; MAY = optional/configurable.

2. Roles and organization

- Super Admin: manages user accounts, warehouse/client assignments, company catalog, approvals, configuration, cross-organization oversight, reports, and audit.
- Store Admin: operates one assigned warehouse; views stock/history; submits stock-in/out, transfer, and correction requests; views/exports authorized warehouse reports. Cannot create products or categories.
- Client Admin: manages its client catalog, manufacturing, department change requests, client inventory workflows, issues/returns, and authorized client reports. Client product transactions are blocked until product approval.
- Warehouses belong to the company, not clients. Super Admin controls a many-to-many client–warehouse assignment. Removing an assignment blocks new requests but does not invalidate approved/in-transit shipments or erase history.
- Client Admin does not see the source warehouse identity for inbound shipments. Only Super Admin creates, disables, or resets accounts. Shared client account handover preserves data; audit identifies the account, not necessarily the human.

3. Catalog and product model

- Two catalogs: company products held in warehouses, managed by Super Admin; and client products manufactured or stocked by a client, managed by Client Admin and visible to Super Admin.

Product fields include name, description, category/subcategory, optional brand/image, stock unit, conversion definitions, configurable physical specifications, batch/expiry tracking settings, decimal precision policy, minimum threshold, active status, and audit metadata. No mandatory SKU field.

- Use common measurement fields plus configurable custom attributes. Stock unit/count is separate from physical dimensions. Explicit unit conversion ratios are versioned; historical transactions preserve the version used.
- Quantity precision is configurable up to two decimal places; whole-unit products accept integers only. Use decimal arithmetic, never binary floating-point for stock calculations.
- Duplicate detection is programmed, not AI-based: exact normalized duplicates are blocked pending resolution; similar names warn. Different sizes/grades/specifications may be distinct products.

Client-created products remain pending until Super Admin approval. Descriptive edits may be low-risk; stock-unit, conversion, or stock-relevant specification changes require controlled review. Deactivation is blocked until stock, reservations, and pending workflows are resolved. Reactivation never restores stock automatically.

4. Inventory states and calculations

- Track inventory by owner/location, product, and where configured batch and condition. Distinguish physical stock, usable stock, reserved stock, available stock, in-transit stock, and pending stock-in.
- Available stock = usable physical stock – active reservations. Expired, damaged, or blocked quantities are not usable. Pending stock-in is not physical or available until approved.
- Example: 100 available; requests for 90 and 10 reserve all 100. Pending stock-in of 50 does not change availability. After approval, eligible incoming stock becomes available while existing reservations remain intact.
- Reserve stock-out quantities atomically at submission. Combined reservations cannot exceed eligible available stock. Approval consumes the reservation without double deduction; rejection/cancellation releases it exactly once.
- Pending edits revalidate atomically. Failed edits preserve the original request and reservation. Reductions release excess. Product changes reserve the replacement before releasing the old reservation. Reservations do not expire automatically.
- No negative stock. Concurrent requests use database locking/conditional updates and retry-safe transactions.

5. Ledger and transaction integrity

- The append-only stock ledger is the authoritative record of finalized movements. Balance tables are read-optimized projections and must reconcile to ledger totals.
- Each movement records unique transaction ID, type, product, quantity, unit, conversion version, source/destination, batch/condition when applicable, actor, approval, timestamps, and source request.

- Movement and balance updates commit in one database transaction or roll back entirely. Approval is idempotent; duplicate clicks/retries cannot duplicate stock movements.
- Historical corrections use linked reversal/correction entries; never silently overwrite finalized records. Physical count adjustments are separate from historical transaction corrections.

Use fixed-precision decimal/numeric types. Reconciliation detects and investigates differences; it never silently rewrites history. Transfers link source, in-transit, and destination stages. Linked manufacturing consumes inputs and creates outputs atomically.

6. Requests and approvals

- Default lifecycle: Draft → Submitted → Pending Approval → Approved or Rejected. Stock-out requests also create reservations. Original requester may cancel their own pending request; cancelled requests cannot later be approved.
- All inventory-affecting operations require Super Admin approval: stock-in/out, opening balances, transfers, manufacturing, returns, write-offs, adjustments, department issues, and corrections.
- Super Admin can approve eligible requests in any order, subject to workflow dependencies. Approval requires a valid session and server-side permission; record approver, time, decision, and reason.
- Rejected requests retain reasons/history. Completed requests cannot be edited directly. Pending requests can be edited only after validation and atomic reservation recalculation.
- Priority options: Low, Normal, High/Urgent; default Normal. Requesters may suggest priority with a reason; Super Admin confirms/changes/removes it. Priority affects visibility only, never permissions or stock rules.
- Approval is click-to-approve without password re-entry, subject to valid session and authorization. High-risk re-authentication can be reconsidered in a later security review.

7. Dispatch, transfers, and receipt discrepancies

Warehouse-to-client: Store Admin submits request for an assigned client → reserve stock → Super Admin approves dispatch → deduct source stock and create in-transit shipment → if receipt confirmation is enabled, Client Admin records received quantity → Super Admin resolves discrepancies and finalizes receipt → client stock is updated according to the approved received quantity.

- Receipt confirmation is configurable by Super Admin. If disabled, the dispatch-finalization policy must be explicitly specified in the detailed state-transition design.
- Warehouse-to-warehouse: request → dispatch approval → source deduction and in-transit tracking → destination receipt entry → Super Admin receipt approval/discrepancy resolution → destination stock addition for approved quantity.
- Client sees inbound/received quantities without source warehouse identity. Each warehouse stock-out batch has one source warehouse; no automatic warehouse allocation.
- If 100 are dispatched and 95 received, record a discrepancy of 5. Super Admin may approve 95, hold, or request correction. Missing-stock write-off requires a separate authorized decision. All stages remain linked and auditable.

8. Client operations, manufacturing, departments

Manufacturing supports linked and separate modes. Linked mode records raw-material consumption, finished output, planned/actual quantities, and wastage; approval applies inputs and outputs atomically. Separate mode submits raw-material stock-out and finished-product stock-in independently.

- Manufacturing must prevent unavailable input consumption, duplicate finalization, and partial inventory updates. Wastage is explicit and reconciles to the production record.
- Client Admin maintains department records, but create/edit/deactivate changes require Super Admin approval. Approved department issues reduce central client stock; no separate department stock balances are maintained.
- Returns should link to original issue when possible and record quantity, unit, condition, reason, and optional attachment. Usable returns restore central stock after approval; damaged/expired returns remain unavailable.
- Client direct stock-in requires a source type (e.g. purchase, opening stock, department return, donation, other) and Super Admin approval. Opening stock supports manual and reviewed bulk entry; approval is required.

9. Batch, expiry, damage, write-off

- Batch and expiry tracking are configurable per product. Batch number is required when batch tracking is enabled; expiry date when expiry tracking is enabled; both when both are enabled.
- Expiry alerts default to 90, 30, and 7 days before expiry, configurable by Super Admin. Expired stock is excluded from usable stock and blocked from routine issue/dispatch, but remains visible in reports.
- Damaged/expired stock is tracked separately and excluded from usable stock. Store Admin submits quantity/reason; Super Admin approves write-off/deduction. No separate quarantine workflow in initial scope. Disposal requires an approved movement.

10. Bulk imports and AI

- CSV, editable tables, images, and scanned documents can prepare product creation, opening stock, stock-in, stock-out, and corrections. Upload alone never changes stock.
- Workflow: upload → extraction/classification and field mapping → validation → editable review → batch submission → Super Admin approval → atomic finalization.
- AI is limited to image/scanned-document classification, extraction, and field mapping in Super Admin portal. AI never approves or directly modifies stock. Programmed logic performs duplicate detection.
- Invalid rows and field-level errors remain in the editable draft; users correct without reuploading. Repeated product/warehouse rows are flagged; users may keep separate or explicitly merge them. Never silently merge.

Tracking fields are required according to product policy. Validate fields, conversions, precision, permissions, availability, duplicates, and scope. One batch ID links rows. Super Admin chooses partial or all-or-nothing behavior where applicable. Pending/rejected rows do not change stock.

- Bulk stock-out rows must use one source warehouse. Unauthorized/out-of-scope rows cannot bypass permissions. Approved movements are applied atomically at the configured approval unit.

11. Daily closing and batches

- Client Admin may submit transactions individually or group daily transactions into a report/batch. Each transaction retains independent status and audit history.
- Prevent duplicate inclusion in multiple active batches. Support selective rejection and whole-batch rejection. Reports distinguish approved, pending, and rejected transactions.
- All-or-nothing batches commit as one database transaction. Partial batches commit only selected approved rows or explicitly approved subgroups. Pending rows never affect finalized balances.

12. Dashboards and notifications

- Super Admin gets a unified approval overview and filtered queues for warehouse movements, client inventory, manufacturing, departments, returns, corrections, and daily batches. In-app notifications and persistent pending-request bar are enabled.
- Requesters receive approval/rejection outcomes. Low-stock alert persists until stock recovers to threshold. Notify when stock first drops below threshold; notify again only after recovery and a later new drop.
- Super Admin sees company-wide and relevant client alerts; Client Admin sees own client alerts; Store Admin receives no low-stock notifications under the agreed policy.
- Sound is configurable and off by default. No email/SMS/push in initial scope. Backup failure triggers retry, Super Admin notification, and persistent dashboard warning until success.

13. Reports and exports

Reports: current stock by warehouse/client/product/condition/batch; physical, usable, reserved, available, pending inbound, and in-transit quantities; stock-in/out history; dispatch/receipt/transfer/discrepancy; low-stock/expiry; opening stock/adjustments; manufacturing input/output/wastage; department issues/returns; request statuses; daily batches; audit/user activity.

- Filters include date range, product, warehouse, client, transaction type, status, and user, subject to permissions. Exports: XLSX, CSV, PDF, print-friendly. Exports are audited. Valuation, costs, prices, and currency are out of scope.

14. Security baseline

- Username/password; no OTP initially. Minimum 12-character passwords, strength checks, and blocking of common/predictable/known-compromised passwords.
- Five consecutive failed logins trigger a 15-minute lockout; repeated lockouts require Super Admin intervention; successful login resets the counter.
- One active session per account; new login invalidates prior session. Seven-day inactivity timeout. Password reset and account disable invalidate sessions immediately.
- HTTPS/TLS; adaptive salted password hashing; secrets never logged. Server-side authorization on every API route, including client/warehouse scope checks.

Private file storage, upload type/size limits, safe download authorization, and malware scanning where available. Add rate limits, secure cookies, CSRF protections where applicable, input validation, output encoding, dependency patching, and monitoring.

- Append-only audit events cover movements, decisions, users, sessions, catalog, assignments, settings, and exports. Shared accounts cannot prove which individual used them; do not claim otherwise.

15. Audit and retention

- Audit events include actor account, action, target, timestamp, outcome, request/transaction reference, and relevant before/after values. Never record passwords, session tokens, or secrets.
- Ordinary application paths must not edit/delete historical audit events. Administrative access and retention operations are controlled and logged.
- 90-day retention applies to backups, not automatically to operational records. Confirm legal/business retention before production. Do not delete history just because a product or account is disabled.

16. Backup, recovery, and operations

- Daily automated backups retained 90 days; encrypt at rest/in transit; restrict access; retry failures and alert Super Admin until success.
- Back up database and required attachments consistently. Schedule restoration tests and record outcomes; a backup is not proven until restored successfully.
- Monitor application errors, database health, storage, failed jobs, backup status, and reconciliation exceptions. Maintain separate development/test, staging, and production environments.
- Before provider selection, define recovery-point objective (maximum acceptable data loss) and recovery-time objective (maximum acceptable downtime).

17. Recommended architecture

Start with a modular monolith: one deployable backend with clear modules for identity/access, catalog, inventory ledger, requests/approvals, logistics, manufacturing, imports, notifications, reporting, and audit. This avoids premature microservice complexity.

Frontend: responsive browser application. Backend: secure API with centralized business rules. Database: PostgreSQL or equivalent relational database with transactions, constraints, and row-level locking. Store files in private object storage. Use background workers for imports, expiry checks, notifications, backups, and scheduled tasks.

- Hosting/provider remains deferred until workload, data residency, budget, support, recovery objectives, and availability requirements are reviewed.

18. Conceptual data model

- Organization/access: User, Role, UserSession, Client, Warehouse, ClientWarehouseAssignment.
- Catalog: Product, Category, ProductAttributeDefinition, ProductAttributeValue, Unit, ProductUnitConversionVersion, ProductTrackingPolicy, ProductMinimumThreshold, Batch.
- Inventory: InventoryBalance, StockReservation, StockLedgerEntry, StockTransaction, StockTransactionLine, Shipment, ShipmentReceipt, Transfer, Discrepancy.
- Client operations: ManufacturingOrder, ManufacturingInput, ManufacturingOutput, Department, DepartmentChangeRequest, DepartmentIssue, DepartmentReturn.
- Governance: ApprovalDecision, AuditEvent, ImportBatch, ImportRow, Notification, ReportExport, BackupRun, ConfigurationSetting.

Relationships: client and warehouse have many-to-many assignment; request has lines; stock-out line may create reservation; finalized transaction creates ledger entries; transfer links source/in-transit/destination; manufacturing links inputs/outputs; import rows belong to a batch.

19. API/workflow contracts

All endpoints authenticate, authorize scope, validate input, and audit protected changes. Required API families: identity/session; organization and assignments; catalog and product approval; inventory balances/availability/ledger; requests/edit/submit/cancel/approve/reject; logistics dispatch/transfer/receipt/discrepancy; manufacturing; department issue/return; bulk upload/draft/validate/edit/revalidate/submit/approve; reports/exports; audit/notifications/backup/reconciliation.

- Use idempotency keys or unique request identifiers for retryable mutations. Return clear validation errors and current request status. Clients must never directly set finalized stock balances.

20. Acceptance and verification tests

Concurrent stock-outs cannot reserve beyond available stock. Pending stock-in never changes availability. Duplicate approval creates one movement only. Cancellation/rejection releases reservation exactly once. Failed approval causes no partial ledger/balance update.

- Receipt discrepancies reconcile dispatched, received, in-transit, and missing quantities. Conversion precision is safe and historical versions stable. Expired/damaged stock cannot be routinely issued.

- Cross-client/warehouse access is denied and logged. Bulk imports remain drafts until approval; invalid rows stay editable; duplicate rows are not silently merged. Partial/all-or-nothing batch behavior matches specification.
- Audit remains traceable after correction, deactivation, and handover. Restore database and attachments from backup and verify integrity. Conduct role-based user acceptance testing.

21. Phased delivery plan

- Phase 1: requirements, state transitions, ERD, permissions, API contracts, threat model, deployment objectives.
- Phase 2: authentication, users, clients, warehouses, assignments, audit foundation.
- Phase 3: catalog, units/conversions, batches, ledger, balances, reservations.
- Phase 4: approvals, stock-in/out, opening balances, transfers, receipts, corrections, write-offs.
- Phase 5: client catalog, manufacturing, departments, issues, returns.
- Phase 6: bulk import/review, AI extraction, dashboards, notifications, reports.
- Phase 7: concurrency/security tests, reconciliation, restoration drills, user acceptance, controlled production rollout.

22. Items to explicitly confirm before production

- Exact receipt workflow when client receipt confirmation is disabled.
- Whether partial/all-or-nothing approval applies to each transaction type and the exact atomicity boundary.
- Final product duplicate key and attribute normalization rules.
- Operational ledger/audit retention period and any regulatory obligations.
- Recovery-point/recovery-time objectives, expected users/transactions, uptime target, data residency, and hosting budget.
- Whether shared client accounts remain acceptable long-term or should later be replaced with named individual accounts.
- No unresolved item is silently assumed during implementation; changes must be documented and assessed for impact on permissions, balances, audit, and existing workflows.